

Mathematics for AI

Basic Algebra, Functions, Scalars, and Vectors

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Learning Outcomes

By the end of this lecture, you should be able to:

- Understand the importance of mathematics in artificial intelligence.
- Explain key concepts of algebra, including functions and equations.
- Describe and work with scalars and vectors in AI contexts.
- Perform basic vector operations and interpret similarity measures.

Why Do We Need Math in AI?

- Math helps describe the **logic**, **structure**, and **learning rules** of AI.
- AI models operate on **numerical data**, using math to find patterns.
- Even models that look like black boxes (like deep neural networks) are built entirely using mathematical operations.
- Key areas we start with are algebra, functions, scalars, and vectors.

Algebra in AI

- Algebra allows us to build mathematical expressions representing models.
- For example: $y=wx+b$
 - y : predicted output
 - x : input data
 - w : model's weight or importance given to input
 - b : bias or base value
- Used in all AI models including linear regression, SVMs, neural networks, etc.
- Learning involves adjusting and to get better predictions.

Real Example: Linear Regression

- **Scenario:** Predict house price based on its size (square feet)
- **Model equation:** $y=wx+b$

Where:

- y : predicted house price
- x : size of the house
- w : learned weight (price per unit area)
- b : bias (baseline price)

- **Prediction example:**

If $w = 200$, $b = 50000$, and $x=1000$: $y = 200 \cdot 1000 + 50000 = 250000$.

- **Error Minimization:**

Use **Mean Squared Error (MSE)** as the cost function to evaluate predictions:

Where:

- y_i : actual value
- $\hat{y}_i$: predicted value
- n : number of training samples

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Multiple Linear Regression

- **Scenario:** Predict price using multiple features: size, number of rooms, location score.
- **Model equation:** $y = w_1x_1 + w_2x_2 + w_3x_3 + b$
- Or in vector form: $y = \vec{w} \cdot \vec{x} + b$
- Each weight w_i represents the contribution of feature x_i
- **Used in** many regression tasks including pricing, forecasting, etc.

What is a Function?

- A **function** is a mathematical rule that assigns to each input exactly one output.

- **Formal definition:**

A function $f : X \rightarrow Y$ maps every element $x \in X$ to a unique element $f(x) \in Y$.

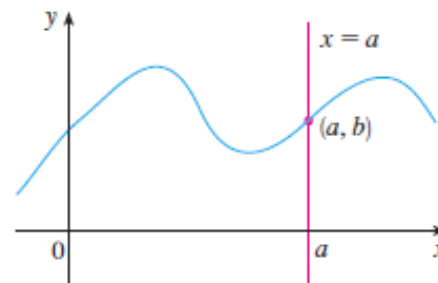
- **Examples:**

$$f(x) = x + 3$$

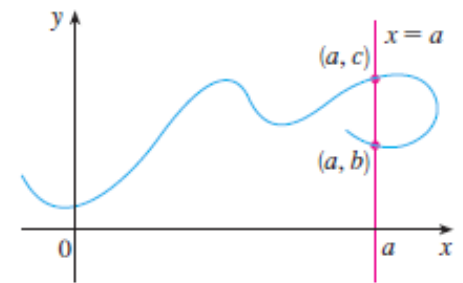
$$f(x) = x^2$$

- **In AI, functions are used to:**

- Transform inputs
- Model relationships
- Activate neurons
- Make decisions



(a) This curve represents a function.

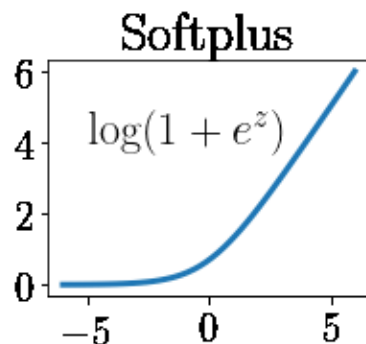
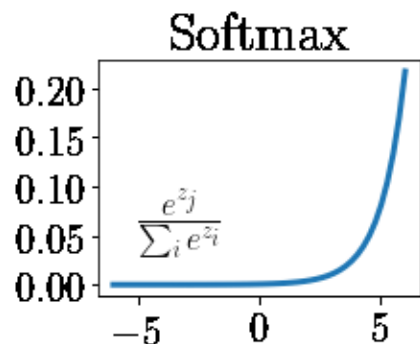
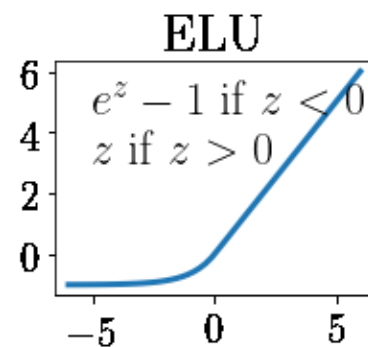
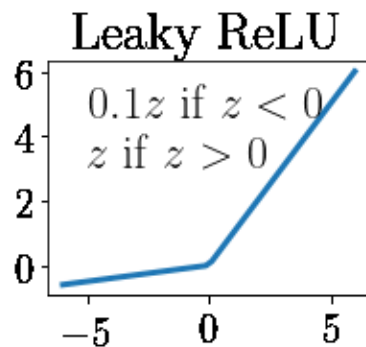
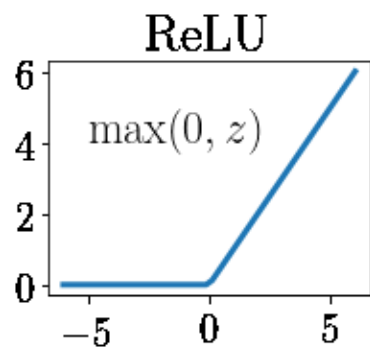
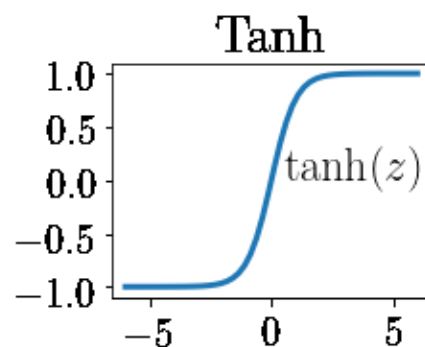
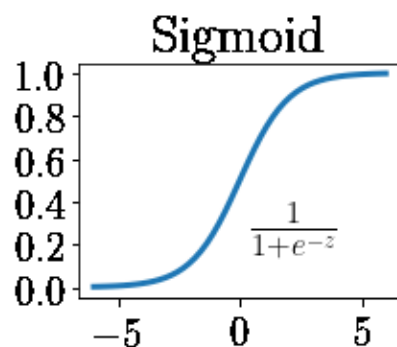
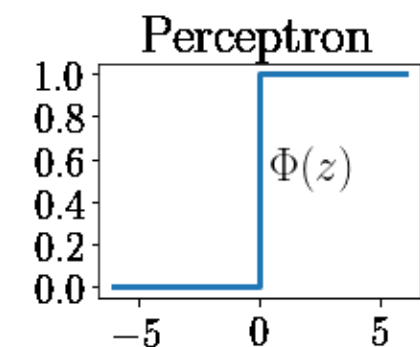


(b) This curve doesn't represent a function.

Several Activation Functions

Function	Formula	Properties
Sigmoid	$\frac{1}{1+e^{-x}}$	Smooth curve, squashes values between 0 and 1
ReLU	$\max(0, x)$	Simple and efficient, used in deep learning
Tanh	$\tanh(x)$	Outputs between -1 and 1
Softmax	$\frac{e^{x_i}}{\sum_j e^{x_j}}$	Used for multi-class classification
Perceptron	$f(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$	Step function, used in early neural networks
Leaky ReLU	$f(x) = \begin{cases} x & \text{if } x \geq 0 \\ \alpha x & \text{if } x < 0 \end{cases}$	Allows small gradient when $x < 0$, helps avoid dying ReLU
ELU	$f(x) = \begin{cases} x & \text{if } x > 0 \\ \alpha(e^x - 1) & \text{if } x \leq 0 \end{cases}$	Smooth output, helps with faster convergence
Softplus	$\ln(1 + e^x)$	Smooth approximation of ReLU, always positive, differentiable everywhere

Graphs of Activation Functions



CNN Functional Overview

1. Convolutional Layer(s)

- Applies filters (kernels) to extract features.
- Output size (per dimension): $\left(\frac{W-F+2P}{S}\right) + 1$ where:
 W : input width or height, F : filter size, P : padding, S : stride
- Each filter produces one 2D feature map.

2. ReLU Activation

Applies non-linearity: $f(x) = \max(0, x)$

3. Pooling Layer

Reduces spatial dimensions using operations like max or average pooling. Common formula (e.g., for max pooling): $\left(\frac{W-F}{S}\right) + 1$ (typically, $P = 0$ for pooling)

4. Flatten Layer

Converts 3D feature maps to a 1D vector of size: $H \times W \times D$

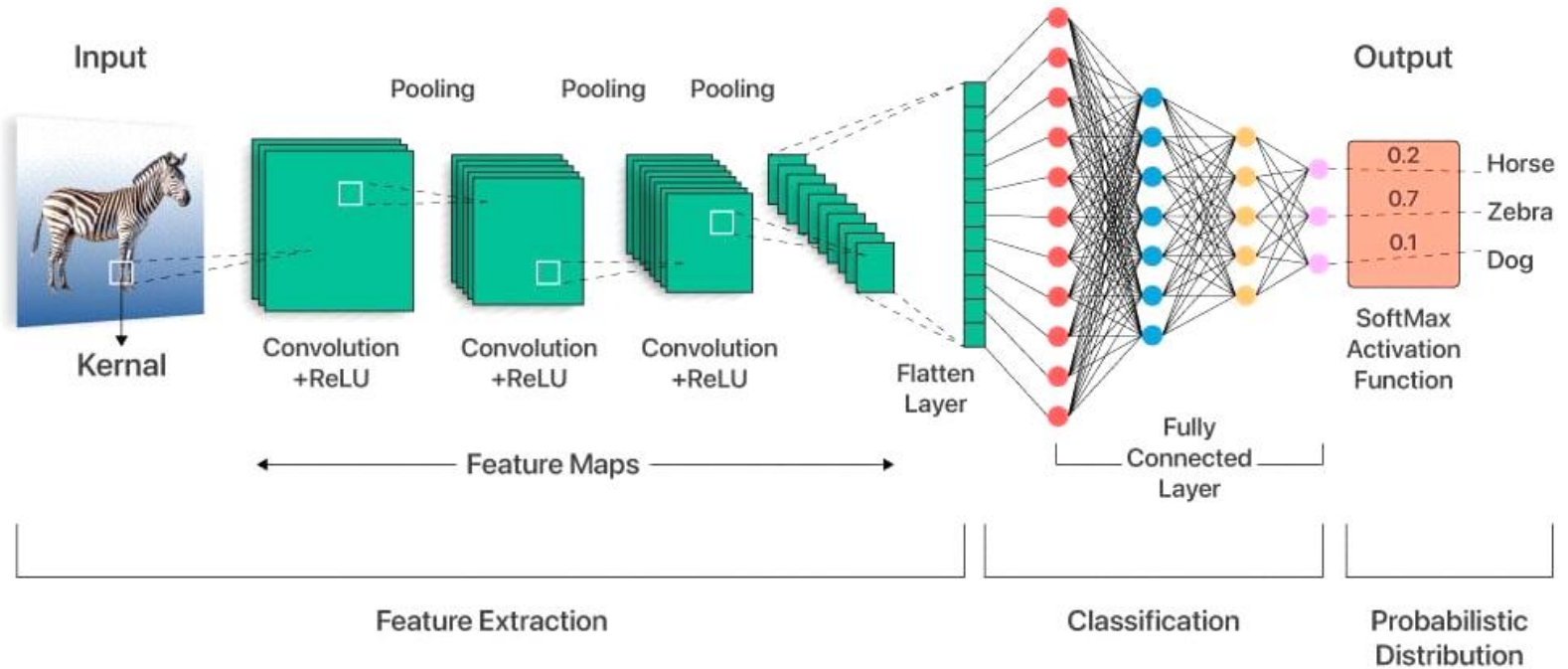
5. Fully Connected (Dense) Layer(s)

- Computes weighted sum of inputs: $z = \sum_{i=1}^n w_i x_i + b$
 x_i : input features, w_i : weights, b : bias
- Output after activation (e.g., ReLU or Softmax): $a = f(z)$

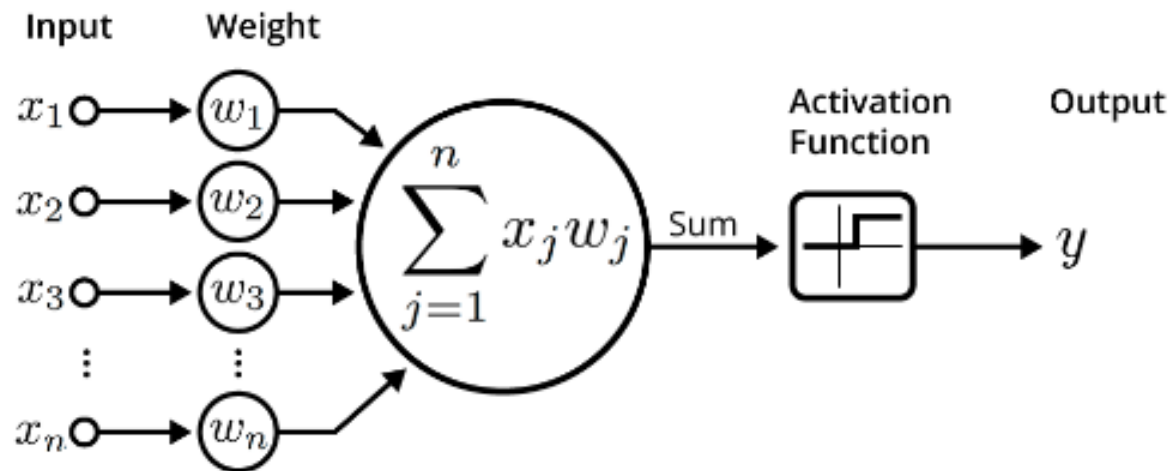
6. Output Layer

Uses softmax for classification: $\text{Softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$

CNN Functional Overview



Neuron in a Neural Network



Scalars

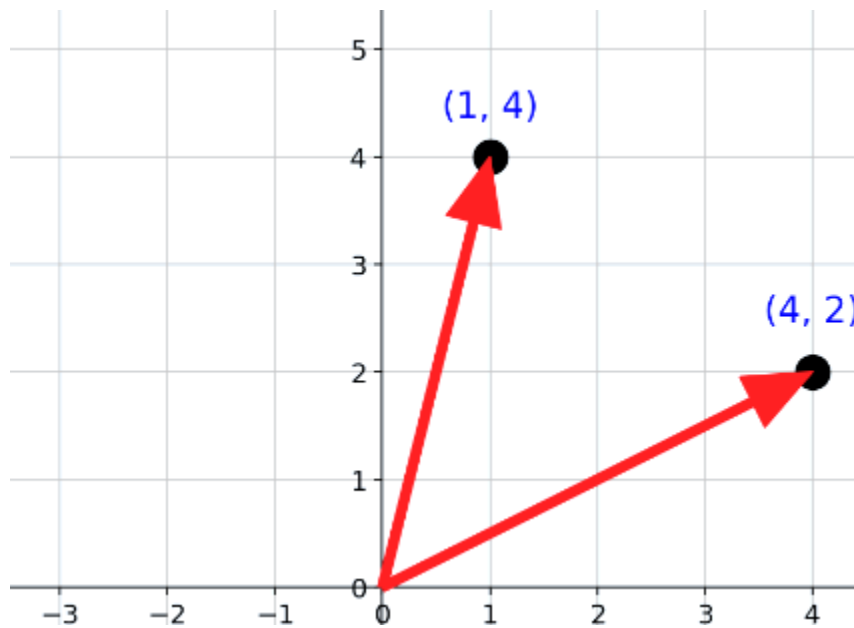
- **Definition:** A scalar is a single numeric value that can represent magnitude but not direction. It belongs to sets like real numbers ($\mathbb{R}$) or integers ($\mathbb{Z}$).
- **Characteristics:** Scalars are used in many basic operations like scaling vectors or defining thresholds. They are dimensionless and have no orientation in space.
- **Examples:**
 - Temperature: 27°C
 - Learning rate in training: 0.01
 - Price of a product: \$12.50
 - A constant weight in a linear model: 1.5
- **Usage in AI:** Scalars are commonly used as weights, biases, loss values, and activation thresholds. They control the scale or influence of vector and matrix operations.

Vectors

- **Definition:** A vector is a mathematical object consisting of an ordered sequence of numbers, used to represent both magnitude and direction. Vectors are foundational in artificial intelligence for encoding data, inputs, and model parameters.
- **Notation:**
 - **Row Vector ($1 \times n$):** A horizontal list of components, written as: $[v_1, v_2, \dots, v_n]$
 - **Column Vector ($n \times 1$):** A vertical arrangement of components, written as: $\begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$
 - If $\mathbf{A}$ is a column vector, then $\mathbf{A}^T$ is the row vector of $\mathbf{A}$ and vice versa.
- **Size/Dimension:**
 - The size of a vector refers to the number of elements it contains. For example, a vector with three components is said to be in $\mathbb{R}^3$.
 - Row and column vectors have different orientations but represent the same data.
- **Examples:**
 - 2D vector as a row: $[4, 3]$
 - 2D vector as a column: $\begin{bmatrix} 4 \\ 3 \end{bmatrix}$
- **Relevance in AI:** Vectors are used throughout AI and ML to represent data points, model weights, and feature embeddings.

Vectors - Graphical Representation in 2D

Vectors $[1,4]$ and $[4,2]$



Vector Operations

- **Vector Addition:** The sum of two vectors is obtained by adding their corresponding components.
 - Example: $[1, 2] + [3, 4] = [4, 6]$
 - Geometric Interpretation: Represents tip-to-tail addition.
- **Scalar Multiplication:** A vector can be scaled by multiplying each of its components by a scalar value.
 - Example: $2 \times [1, 3] = [2, 6]$
 - Application: Used to stretch or shrink vectors.
- **Dot Product (Inner Product):** Measures the extent to which two vectors point in the same direction.
 - Formula: $a \cdot b = a_1b_1 + a_2b_2 + \dots + a_nb_n$
 - Example: $[1, 2] \cdot [3, 1] = 1 \times 3 + 2 \times 1 = 5$
 - Applications: Similarity calculations, projection, and determining orthogonality.
- **Vector Subtraction:** Subtracts each corresponding component.
 - Example: $[5, 7] - [2, 3] = [3, 4]$
 - Interpretation: Directional difference between vectors.
- **Relevance in AI:** These operations are foundational for understanding geometric relationships in high-dimensional space, manipulating model weights, computing gradients, and performing similarity-based tasks such as in recommendation systems and clustering

Unit Vectors and Direction

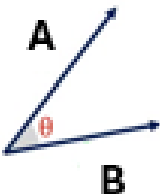
- **Definition:** A unit vector is a vector that has a magnitude (length) of exactly 1. It retains the direction of the original vector but has a standardized scale.
- **Notation:** A unit vector is often denoted by a hat symbol, such as $\hat{v} = v / \|v\|$, where v is any non-zero vector and $\|v\|$ is its norm.
- **Purpose and Interpretation:**
 - Used to represent **pure direction** without concern for magnitude.
 - All unit vectors lie on the surface of a unit circle (2D) or unit sphere (3D or higher).
- **Example:**
 - Given $v = [3, 4]$, its unit vector is:
 - $\|v\| = \sqrt{3^2 + 4^2} = 5$
 - $\hat{v} = [3/5, 4/5] = [0.6, 0.8]$
- **Applications in AI:**
 - In gradient-based learning, unit vectors are used to indicate direction of steepest descent.
 - In normalization of feature embeddings to ensure fair comparisons.
 - In vector-based search systems to simplify similarity computation

Vector Length and Normalization

- **Vector Norm (or Length):** Measures the magnitude of a vector — the distance from the origin to the point represented by the vector.
 - The most common norm is the **Euclidean norm (L2 norm)**: $\|v\|_2 = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$
 - Other norms include:
 - **L1 norm:** $\|v\|_1 = |v_1| + |v_2| + \dots + |v_n|$
 - **Infinity norm:** $\|v\|_\infty = \max(|v_1|, \dots, |v_n|)$
- **Normalization:** The process of converting a vector to a **unit vector** (length 1) while preserving its direction.
 - Formula: $\hat{v} = v / \|v\|$
 - Ensures all vectors lie on the unit hypersphere in high-dimensional space.
- **Why Normalize Vectors?**
 - Prevents features with large values from dominating model behavior.
 - Essential for calculating cosine similarity and direction-based comparisons.
 - Common in neural networks, word embeddings, and recommender systems.
- **Example:**
 - Let $v = [3, 4]$, then $\|v\| = \sqrt{3^2 + 4^2} = 5$
 - Normalized vector: $\hat{v} = [3/5, 4/5] = [0.6, 0.8]$

Cosine Similarity

- **Definition:** Cosine similarity measures the cosine of the angle between two non-zero vectors. It reflects how similar the directions of the two vectors are, regardless of their magnitudes.
- **Formula:** $\cos(\theta) = (A \cdot B) / (||A|| \times ||B||)$
 - $A \cdot B$ is the dot product of vectors A and B.
 - $||A||$ and $||B||$ are the magnitudes (L2 norms) of vectors A and B.
- **Interpretation:**
 - Range: -1 to 1
 - 1 $\rightarrow$ perfectly aligned (maximum similarity)
 - 0 $\rightarrow$ orthogonal (no similarity)
 - -1 $\rightarrow$ opposite directions (maximum dissimilarity)
- **Example Calculation:**
 - Let $A = [1, 2]$, $B = [3, 4]$
 - $A \cdot B = 1 \times 3 + 2 \times 4 = 11$
 - $||A|| = \sqrt{1^2 + 2^2} = \sqrt{5}$, $||B|| = \sqrt{3^2 + 4^2} = 5$
 - Cosine similarity = $11 / (\sqrt{5} \times 5)$


$$\cos \theta = \frac{A \cdot B}{||A|| \times ||B||}$$

Practice Problems

- 1. Function Evaluation:** Evaluate $f(-2)$, where $f(x) = 3x^2 - 2x + 4$.
- 2. Vector Representation:** Convert the row vector $[1, -2, 3]$ into column form and identify its dimension.
- 3. Dot Product:** Compute the dot product of vectors $a = [1, 2, 3]$ and $b = [4, 0, -1]$.
- 4. Vector Norm and Unit Vector:** For $v = [6, 8]$, find the Euclidean norm and the unit vector.
- 5. Cosine Similarity:** Calculate the cosine similarity between vectors $A = [1, 0]$ and $B = [0, 1]$.
- 6. Simple Linear Regression:** Find the equation of the line passing through the points $(2, 4)$ and $(4, 10)$.
- 7. Multiple Linear Regression:** Given the model $y = \beta_0 + \beta_1x_1 + \beta_2x_2$, and values $\beta_0 = 2$, $\beta_1 = 3$, $\beta_2 = -1$, $x_1 = 4$, $x_2 = 5$, compute y .
- 8. Neuron in NN:** Consider a single neuron in a neural network that receives three inputs: $x_1=1$, $x_2 = -2$, and $x_3=3$. The corresponding weights are $w_1=0.4$, $w_2=-0.6$, and $w_3=0.8$, and the bias term is $b=0.5$. Knowing that the neuron uses a **ReLU** activation function, calculate the value of the neuron.

Answers to Practice Problems

1. **Function Evaluation:** $f(-2) = 3(-2)^2 - 2 \times (-2) + 4 = 3 \times 4 + 4 + 4 = 12 + 4 + 4 = 20$

2. **Vector Representation:** Column vector = $\begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$. Dimension = 3×1 .

3. **Dot Product:** $[1, 2, 3] \cdot [4, 0, -1] = (1 \times 4) + (2 \times 0) + (3 \times -1) = 4 + 0 - 3 = 1$

4. **Vector Norm and Unit Vector:** $\|v\|_2 = \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100} = 10$.
 $\hat{v} = [6/10, 8/10] = [0.6, 0.8]$

5. **Cosine Similarity:** Dot product = $1 \times 0 + 0 \times 1 = 0$. Norms = $\|[1, 0]\| = 1$, $\|[0, 1]\| = 1$.
 $\cos(\theta) = 0 / (1 \times 1) = 0$.

6. **Simple Linear Regression:** Slope $m = (10 - 4)/(4 - 2) = 6/2 = 3$.
Intercept $b = 4 - 3 \times 2 = -2$ Equation: $y = 3x - 2$

7. **Multiple Linear Regression:** $y = 2 + 3 \times 4 - 1 \times 5 = 2 + 12 - 5 = 9$

8. $z = w_1x_1 + w_2x_2 + w_3x_3 + b$

$$z = (0.4)(1) + (-0.6)(-2) + (0.8)(3) + 0.5$$

$$z = 0.4 + 1.2 + 2.4 + 0.5 = 4.5$$

$$\text{ReLU}(z) = \max(0, 4.5) = 4.5$$

Closing Remarks

- This lecture introduced the mathematical foundations essential for AI, including core algebraic operations, scalar and vector concepts, and their practical relevance.
- These tools are the basis for more complex mathematical frameworks used in machine learning and deep learning.
- In the next lecture, we will explore **vector spaces** and **matrix operations**.
- "Mathematics is not just a tool—it is the language through which AI understands the world."